CHAPTER 1

A Brief History



The garden pea plant. This was the experimental subject of Gregor Mendel's genetic investigation. *Holt Studios International (Nigel Cattlin)/Photo Researchers, Inc.*

What is molecular biology? The term has more than one definition. Some define it very broadly as the attempt to understand biological phenomena in molecular terms. But this definition makes molecular biology difficult to distinguish from another well-known discipline, biochemistry. Another definition is more restrictive and therefore more useful: the study of gene structure and function at the molecular level. This attempt to explain genes and their activities in molecular terms is the subject matter of this book.

Molecular biology grew out of the disciplines of genetics and biochemistry. In this chapter we will review the major early developments in the history of this hybrid discipline, beginning with the earliest genetic experiments performed by Gregor Mendel in the mid-19th century. In Chapters 2 and 3 we will add more substance to this brief outline. By definition, the early work on genes cannot be considered molecular biology, or even molecular genetics, because early geneticists did not know the molecular nature of genes. Instead, we call it transmission

genetics **because it deals with the transmission of traits from parental organisms to their offspring. In fact, the chemical composition of genes was not known until 1944. At that point, it became possible to study genes as molecules, and the discipline of molecular biology was born.** ■

1.1 Transmission Genetics

In 1865, Gregor Mendel (Figure 1.1) published his findings on the inheritance of seven different traits in the garden pea. Before Mendel's research, scientists thought inheritance occurred through a blending of each trait of the parents in the offspring. Mendel concluded instead that inheritance is particulate. That is, each parent contributes particles, or genetic units, to the offspring. We now call these particles **genes.** Furthermore, by carefully counting the number of progeny plants having a given **phenotype,** or observable characteristic (e.g., yellow seeds, white flowers), Mendel was able to make some important generalizations. The word *phenotype,* by the way, comes from the same Greek root as *phenomenon,* meaning *appearance.* Thus, a tall pea plant exhibits the tall phenotype, or appearance. *Phenotype* can also refer to the whole set of observable characteristics of an organism.

Mendel's Laws of Inheritance

Mendel saw that a gene can exist in different forms called **alleles.** For example, the pea can have either yellow or green seeds. One allele of the gene for seed color gives rise to yellow seeds, the other to green. Moreover, one allele can be **dominant** over the other, **recessive,** allele. Mendel demonstrated that the allele for yellow seeds was dominant when he mated a green-seeded pea with a yellow-seeded pea. All of the progeny in the first filial generation (F_1) had yellow seeds. However, when these F_1 yellow peas were allowed to self-fertilize, some green-seeded peas reappeared. The ratio of yellow to green seeds in the second filial generation (F_2) was very close to 3:1.

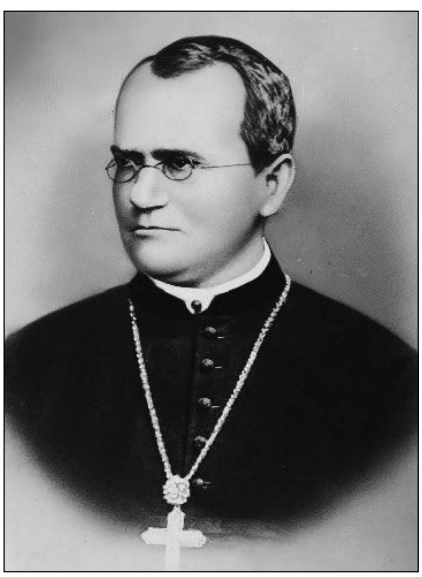


Figure 1.1 Gregor Mendel. (*Source:* Courtesy of Dept. of Library Services/American Museum of Natural History, Neg. no. 219467.)

The term *filial* comes from the Latin: *filius,* meaning son; *filia,* meaning daughter. Therefore, the first filial generation (F_1) contains the offspring (sons and daughters) of the original parents. The second filial generation (F_2) is the offspring of the F_1 individuals.

Mendel concluded that the allele for green seeds must have been preserved in the F_1 generation, even though it did not affect the seed color of those peas. His explanation was that each parent plant carried two copies of the gene; that is, the parents were **diploid,** at least for the characteristics he was studying. According to this concept, **homozygotes** have two copies of the same allele, either two alleles for yellow seeds or two alleles for green seeds. **Heterozygotes** have one copy of each allele. The two parents in the first mating were homozygotes; the resulting F_1 peas were all heterozygotes. Further, Mendel reasoned that sex cells contain only one copy of the gene; that is, they are **haploid.** Homozygotes can therefore produce sex cells, or **gametes,** that have only one allele, but heterozygotes can produce gametes having either allele.

This is what happened in the matings of yellow with green peas: The yellow parent contributed a gamete with a gene for yellow seeds; the green parent, a gamete with a gene for green seeds. Therefore, all the F_1 peas got one allele for yellow seeds and one allele for green seeds. They had not lost the allele for green seeds at all, but because yellow is dominant, all the seeds were yellow. However, when these heterozygous peas were self-fertilized, they produced gametes containing alleles for yellow and green color in equal numbers, and this allowed the green phenotype to reappear.

Here is how that happened. Assume that we have two sacks, each containing equal numbers of green and yellow marbles. If we take one marble at a time out of one sack and pair it with a marble from the other sack, we will wind up with the following results: ¼ of the pairs will be yellow/yellow; ¼ will be green/green; and the remaining ½ will be yellow/green. The alleles for yellow and green peas work the same way. Recalling that yellow is dominant, you can see that only ¼ of the progeny (the green/green ones) will be green. The other ¾ will be *yellow* because they have at least one allele for yellow seeds. Hence, the ratio of yellow to green peas in the second (F_2) generation is 3:1.

Mendel also found that the genes for the seven different characteristics he chose to study operate indepen-

dently of one another. Therefore, combinations of alleles of two different genes (e.g. yellow or green peas with round or wrinkled seeds, where yellow and round are dominant and green and wrinkled are recessive) gave ratios of 9:3:3:1 for yellow/round, yellow/wrinkled, green/round, and green/wrinkled, respectively. Inheritance that follows the simple laws that Mendel discovered can be called **Mendelian inheritance.**

SUMMARY Genes can exist in several different forms, or alleles. One allele can be dominant over another, so heterozygotes having two different alleles of one gene will generally exhibit the characteristic dictated by the dominant allele. The recessive allele is not lost; it can still exert its influence when paired with another recessive allele in a homozygote.

The Chromosome Theory of Inheritance

Other scientists either did not know about or uniformly ignored the implications of Mendel's work until 1900 when three botanists, who had arrived at similar conclusions independently, rediscovered it. After 1900, most geneticists accepted the particulate nature of genes, and the field of genetics began to blossom. One factor that made it easier for geneticists to accept Mendel's ideas was a growing understanding of the nature of chromosomes, which had begun in the latter half of the 19th century. Mendel had predicted that gametes would contain only one allele of each gene instead of two. If chromosomes carry the genes, their numbers should also be reduced by half in the gametes—and they are. Chromosomes therefore appeared to be the discrete physical entities that carry the genes.

This notion that chromosomes carry genes is the **chromosome theory of inheritance.** It was a crucial new step in genetic thinking. No longer were genes disembodied factors; now they were observable objects in the cell nucleus. (See Figure B1.1 to review key structures in the cell.) Some geneticists, particularly Thomas Hunt Morgan (Figure 1.2), remained skeptical of this idea. Ironically, in 1910 Morgan himself provided the first definitive evidence for the chromosome theory.

Morgan worked with the fruit fly *(Drosophila melanogaster),* which was in many respects a much more convenient organism than the garden pea for genetic studies because of its small size, short generation time, and large number of offspring. When he mated red-eyed flies (dominant) with white-eyed flies (recessive), most, but not all, of the F_1 progeny were red-eyed. Furthermore, when Morgan mated the red-eyed males of the F_1 generation with their red-eyed sisters, they produced about ¼ white-eyed males, but no white-eyed females. In other words, the eye color phenotype was **sex-linked.** It was transmitted along with sex in these experiments. How could this be?

We now realize that sex and eye color are transmitted together because the genes governing these characteristics are located on the same chromosome—the X chromosome. (Most chromosomes, called **autosomes,** occur in pairs in a given individual, but the X chromosome is an example of a **sex chromosome,** of which the female fly has two copies and the male has one.) However, Morgan was reluctant to draw this conclusion until he observed the same sex linkage with two more phenotypes, miniature wing and yellow body, also in 1910. That was enough to convince him of the validity of the chromosome theory of inheritance. (See Figure B1.2 to review the cell cycle and the process of mitosis, which explains how traits are transmitted from parent to daughter cells.)

Before we leave this topic, let us make two crucial points. First, every gene has its place, or **locus,** on a chromosome. Figure 1.3 depicts a hypothetical chromosome and the positions of three of its genes, called *A, B,* and *C.* Second, diploid organisms such as human beings normally have two copies of all chromosomes (except sex chromosomes). That means that they have two copies of most genes, and that these copies can be the same alleles, in which case the organism is **homozygous,** or different alleles, in which case it is **heterozygous.** For example, Figure 1.3b shows a diploid pair of chromosomes with different alleles at one locus *(Aa)* and the same alleles at the other two loci (*BB* and *cc*). The **genotype,** or allelic constitution, of this organism with respect to these three genes, is *AaBBcc.* Because this organism has two different alleles (*A* and *a*) in its two chromosomes at the *A* locus, it



Figure 1.2 Thomas Hunt Morgan. (*Source:* National Library of Medicine.)

BOX 1.1

Cell Structure

Before we proceed with our discussion of transmission genetics and molecular genetics, it is important to review cellular structures that are of particular interest in the study of molecular biology (Figure B1.1).

The **nucleus** is a discrete structure within the cell that contains the genetic material, DNA. The DNA is usually present in the cell as a network of fibers called **chromatin.** During mitosis, the DNA molecules form the condensed chromosome structure by coiling and supercoiling around specialized proteins.

Other structures of importance to molecular biologists are the centrosome, the endoplasmic reticulum, the Golgi apparatus, ribosomes, and the mitochondrion. The **centrosome** is a structure composed largely of microtubules. Paired centrosomes organize the formation of spindle fibers during mitosis. The **endoplasmic reticulum** (**ER**) is a network of membranes throughout the cell; cells have both smooth and rough ER. The rough ER is studded with ribosomes, giving it a rough appearance. The bound ribosomes of the ER synthesize proteins destined for secretion from the cell or that are to be incorporated into the membrane or into specific vacuoles. The **Golgi apparatus** is a continuation of the membrane network of the endoplasmic reticulum. **Ribosomes** that are free in the cytosol synthesize proteins that remain in the cell and are not transported through the endoplasmic reticulum and Golgi apparatus. The **mitochondrion** is the site of energy production in the cell. The mitochondrion consists of a double membrane system. The outer membrane is smooth and the inner membrane has convoluted folds. The mitochondrion can divide independently of the cell and contains its own circular, double-stranded DNA.

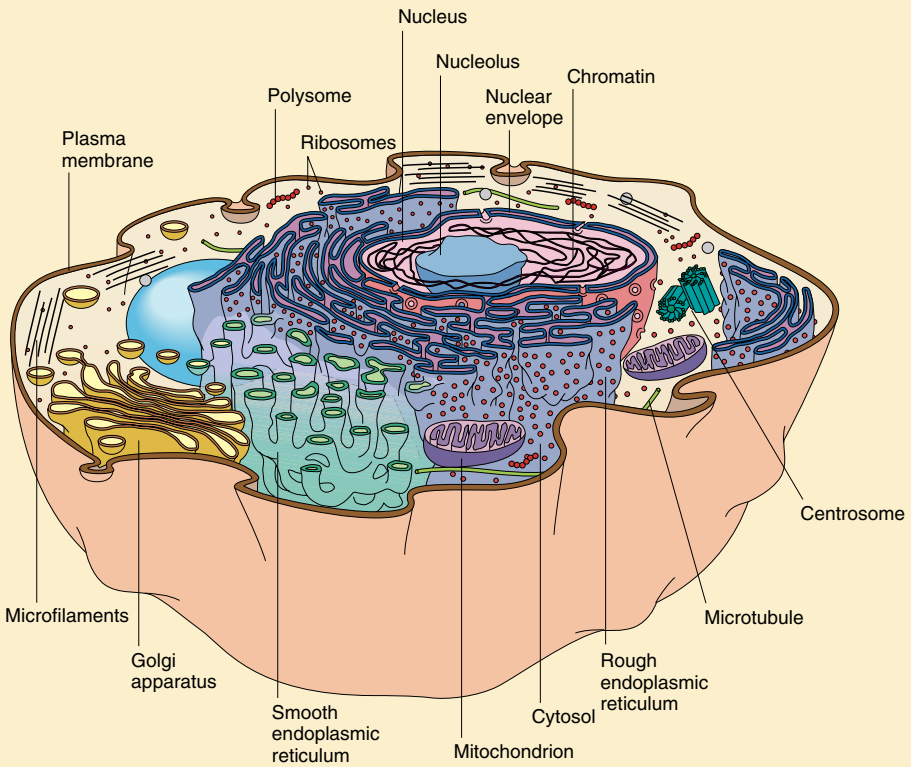


Figure B1.1 Cell structure. Generalized representation of an animal (eukaryotic) cell showing internal structures.

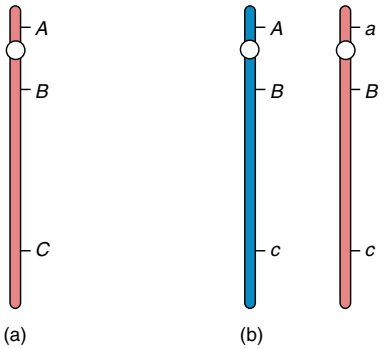


Figure 1.3 Location of genes on chromosomes. (**a**) A schematic diagram of a chromosome, indicating the positions of three genes: *A, B,* and *C.* (**b**) A schematic diagram of a diploid pair of chromosomes, indicating the positions of the three genes—*A, B,* and *C*—on each, and the genotype (*A* or *a; B* or *b;* and *C* or *c*) at each locus.

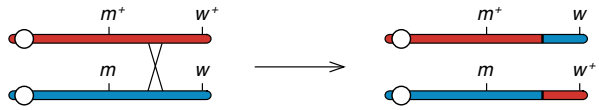


Figure 1.4 Recombination in *Drosophila.* The two X chromosomes of the female are shown schematically. One of them (red) carries two wild-type genes: (m^+), which results in normal wings, and (w^+), which gives red eyes. The other (blue) carries two mutant genes: *miniature* (*m*) and *white* (*w*). During egg formation, a recombination, or crossing over, indicated by the crossed lines, occurs between these two genes on the two chromosomes. The result is two recombinant chromosomes with mixtures of the two parental genes. One is m^+ *w,* the other is *m* w^+.

is heterozygous at that locus (Greek: *hetero,* meaning different). Since it has the same, dominant *B* allele in both chromosomes at the *B* locus, it is homozygous dominant at that locus (Greek: *homo,* meaning same). And because it has the same, recessive *c* allele in both chromosomes at the *C* locus, it is homozygous recessive there. Finally, because the *A* allele is dominant over the *a* allele, the phenotype of this organism would be the dominant phenotype at the *A* and *B* loci and the recessive phenotype at the *C* locus.

This discussion of varying phenotypes in *Drosophila* gives us an opportunity to introduce another important genetic concept: **wild-type** versus **mutant.** The wild-type phenotype is the most common, or at least the generally accepted standard, phenotype of an organism. To avoid the mistaken impression that a wild organism is automatically a wild-type, some geneticists prefer the term **standard type.** In *Drosophila,* red eyes and full-size wings are wild-type. Mutations in the *white* and *miniature* genes result in mutant flies with white eyes and miniature wings, respectively. Mutant alleles are usually recessive, as in these two examples, but not always.

Genetic Recombination and Mapping

It is easy to understand that genes on separate chromosomes behave independently in genetic experiments, and that genes on the same chromosome—like the genes for miniature wing (*miniature*) and white eye (*white*)—behave as if they are linked. However, genes on the same chromosome usually do not show perfect **genetic linkage.** In fact, Morgan discovered this phenomenon when he examined the behavior of the sex-linked genes he had found. For example, although *white* and *miniature* are both on the X chromosome, they remain linked in offspring only 65.5% of the time. The other offspring have a new combination of alleles not seen in the parents and are therefore called **recombinants.**

How are these recombinants produced? The answer was already apparent by 1910, because microscopic examination of chromosomes during meiosis (gamete formation) had shown crossing over between **homologous** chromosomes (chromosomes carrying the same genes, or alleles of the same genes). This resulted in the exchange of genes between the two homologous chromosomes. In the previous example, during formation of eggs in the female, an X chromosome bearing the *white* and *miniature* alleles experienced crossing over with a chromosome bearing the red eye and normal wing alleles (Figure 1.4). Because the crossing-over event occurred between these two genes, it brought together the *white* and normal wing alleles on one chromosome and the red (normal eye) and *miniature* alleles on the other. Because it produced a new combination of alleles, we call this process **recombination.** (See Figure B1.3 to review the process of meiosis and the occurrence of recombination.)

Morgan assumed that genes are arranged in a linear fashion on chromosomes, like beads on a string. This, together with his awareness of recombination, led him to propose that the farther apart two genes are on a chromosome, the more likely they are to recombine. This makes sense because there is simply more room between widely spaced genes for crossing over to occur. A. H. Sturtevant extended this hypothesis to predict that a mathematical relationship exists between the distance separating two genes on a chromosome and the frequency of recombination between these two genes. Sturtevant collected data on recombination in the fruit fly that supported his hypothesis. This established the rationale for **genetic mapping** techniques still in use today. Simply stated, if two loci recombine with a frequency of 1%, we say that they are separated by a map distance of one **centimorgan** (named for Morgan himself). By the 1930s, other investigators found that the same rules applied to other **eukaryotes** (nucleus-containing organisms), including the mold *Neurospora,* the garden pea, maize (corn), and even human beings. These rules also apply to **prokaryotes,** organisms in which the genetic material is not confined to a nuclear compartment.

BOX 1.2

Cell Cycle and Mitosis

Cell Cycle

Cellular reproduction is a cyclic process in which daughter cells are produced through nuclear division (mitosis) and cellular division (cytokinesis). Mitosis and cytokinesis are part of the growth–division cycle called the **cell cycle** (Figure B1.2a). Mitosis, which is divided into four stages (Figure B1.2b), is a relatively small part of the total cell cycle; the majority of the time, a cell is in a growth stage called **interphase.** Interphase is divided into three parts: G_1, S, and G_2. The first gap phase, G_1, follows mitosis and is a period of growth and metabolic activity. The S phase follows G_1 and is a period of DNA synthesis, in which the DNA is replicated. Another gap phase, G_2, follows DNA synthesis and precedes the next mitotic division. Certain mature cell types do not continue to divide but remain in interphase (in G_0).

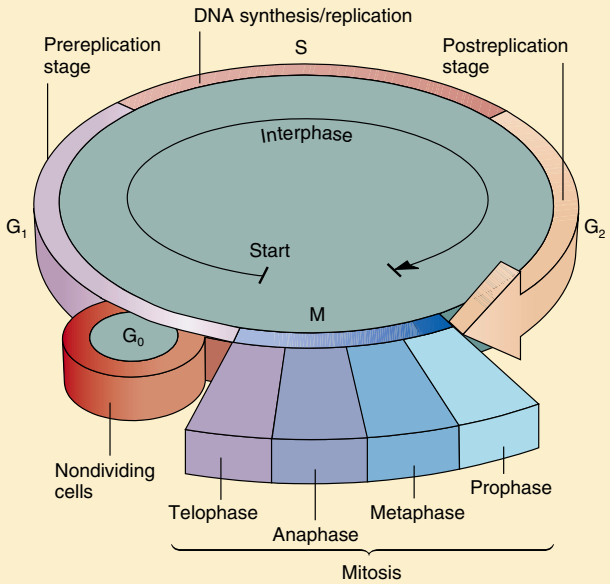


Figure B1.2a Cell cycle. Phases in the cell cycle of a typical eukaryotic cell. The events that occur from the start of one cell division (mitosis), when two daughter cells are produced, to the next cell division are collectively termed the **cell cycle.**

Mitosis

Mitosis is the period in the cell cycle during which the nucleus divides and gives rise to two daughter cells that have chromosome numbers identical to that of the original cell (see Figure B1.2b). Mitosis can be divided into four distinct stages (**prophase, metaphase, anaphase,** and **telophase**) characterized by the appearance and orientation of the homologous pairs of chromosomes.

Prophase

At the end of the G_2 interphase, the chromosomes have replicated, and each has two sister chromatids attached to each other at the centromere. Prophase indicates the beginning of mitosis, when the replicated chromosomes coil and condense. The centrosomes divide and the spindle apparatus also appears.

Metaphase

During metaphase, spindle fibers from the spindle apparatus attach to the centromeres of the chromosomes. The chromosomes gradually migrate to the midline of the cell, oriented between the two centrosomes.

Anaphase

Anaphase begins with the separation of the centromeres and the contraction of the spindle fibers pulling the centromeres toward the centrosomes. As a result, the sister chromatids, now called daughter chromosomes, separate, and the chromosomes move in opposite directions toward the centrosomes.

Telophase

The two sets of chromosomes reach opposite poles of the cell, where they begin to uncoil. Telophase ends with cytokinesis (cellular division) and the formation of two daughter cells.

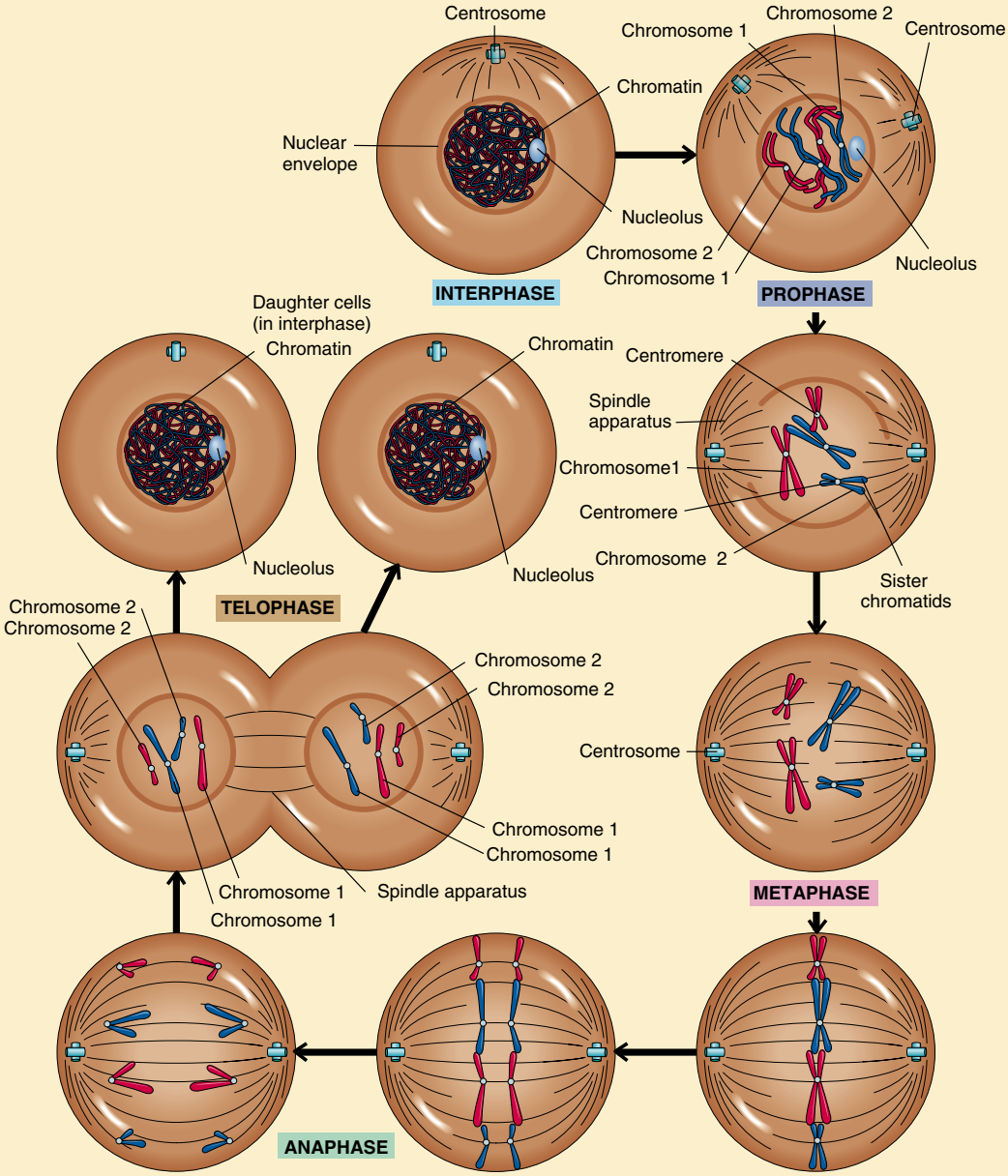


Figure B1.2b **Mitosis.** Mitosis in an animal cell with two homologous pairs of chromosomes.

BOX 1.3

Meiosis

The process of meiosis involves two cell divisions that result in daughter cells (gametes) with half the parental number of chromosomes. If the number of chromosomes were not reduced prior to fertilization, the chromosome number would double in each generation.

Meiosis also allows the exchange of genetic material on homologous chromosomes through recombination, which occurs at the end of prophase I. Recombination results in new combinations of alleles of different genes on a chromosome.

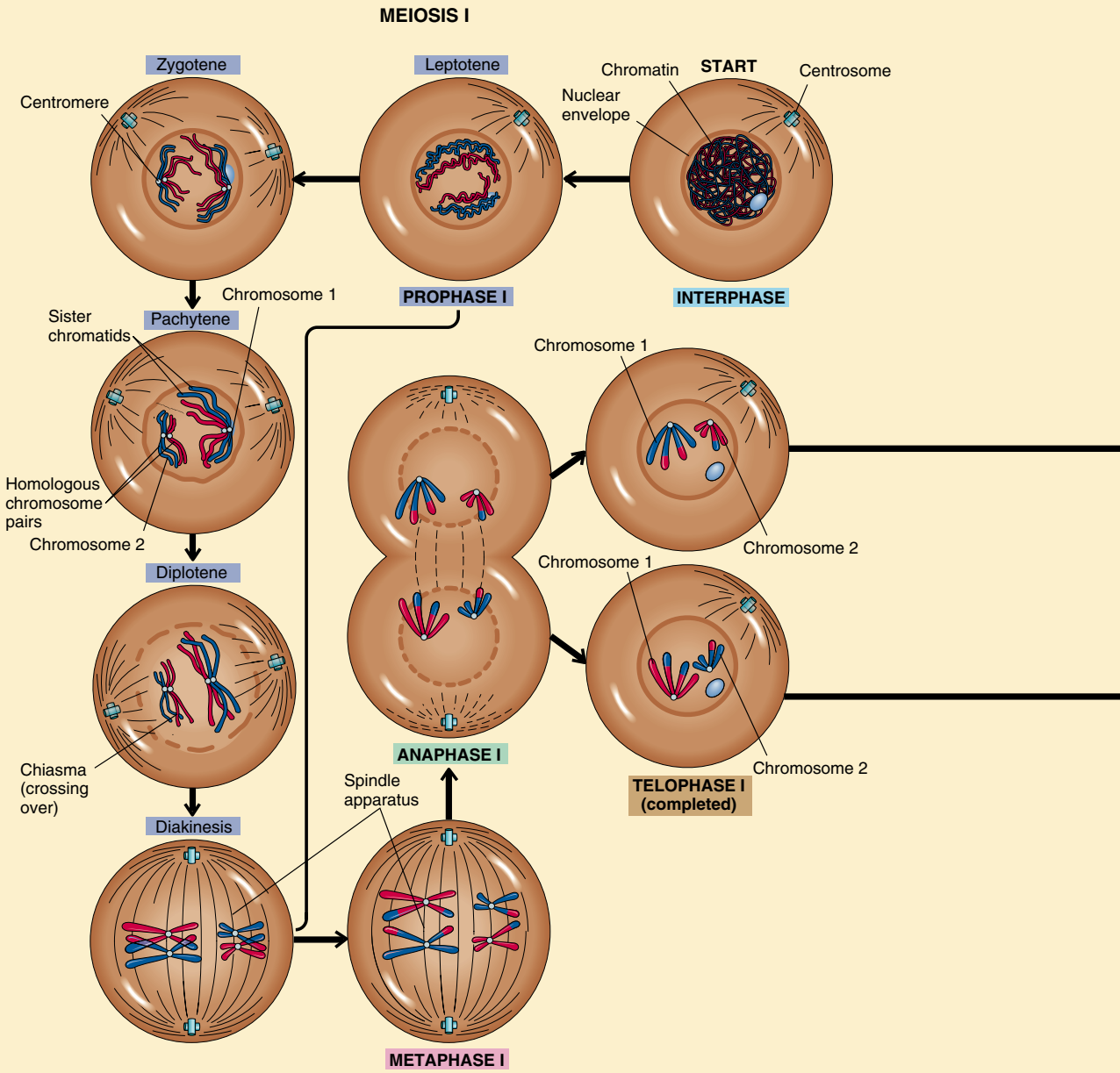


Figure B1.3 Meiosis. A diagrammatic representation of meiosis in a cell with two chromosomes.

Meiosis, like mitosis, can be divided into distinct phases (Figure B1.3). **Meiosis I** includes a multipart prophase I (during which crossing over occurs), metaphase I, anaphase I (unlike in mitosis, homologous pairs of chromosomes, not sister chromatids, separate to the opposite poles of the cell), and telophase I. **Meiosis II** begins with prophase II, continues on to metaphase II (the centromeres attach to the spindle apparatus), anaphase II (as in mitosis, the centromeres separate and the chromosomes begin to migrate to opposite poles), and telophase II, which completes meiosis, resulting in four daughter cells with half the number of chromosomes as the parental cell. As shown in Figure B1.3, the parental cell contains four chromosomes (two homologous pairs), and the resulting four daughter cells have two chromosomes each.

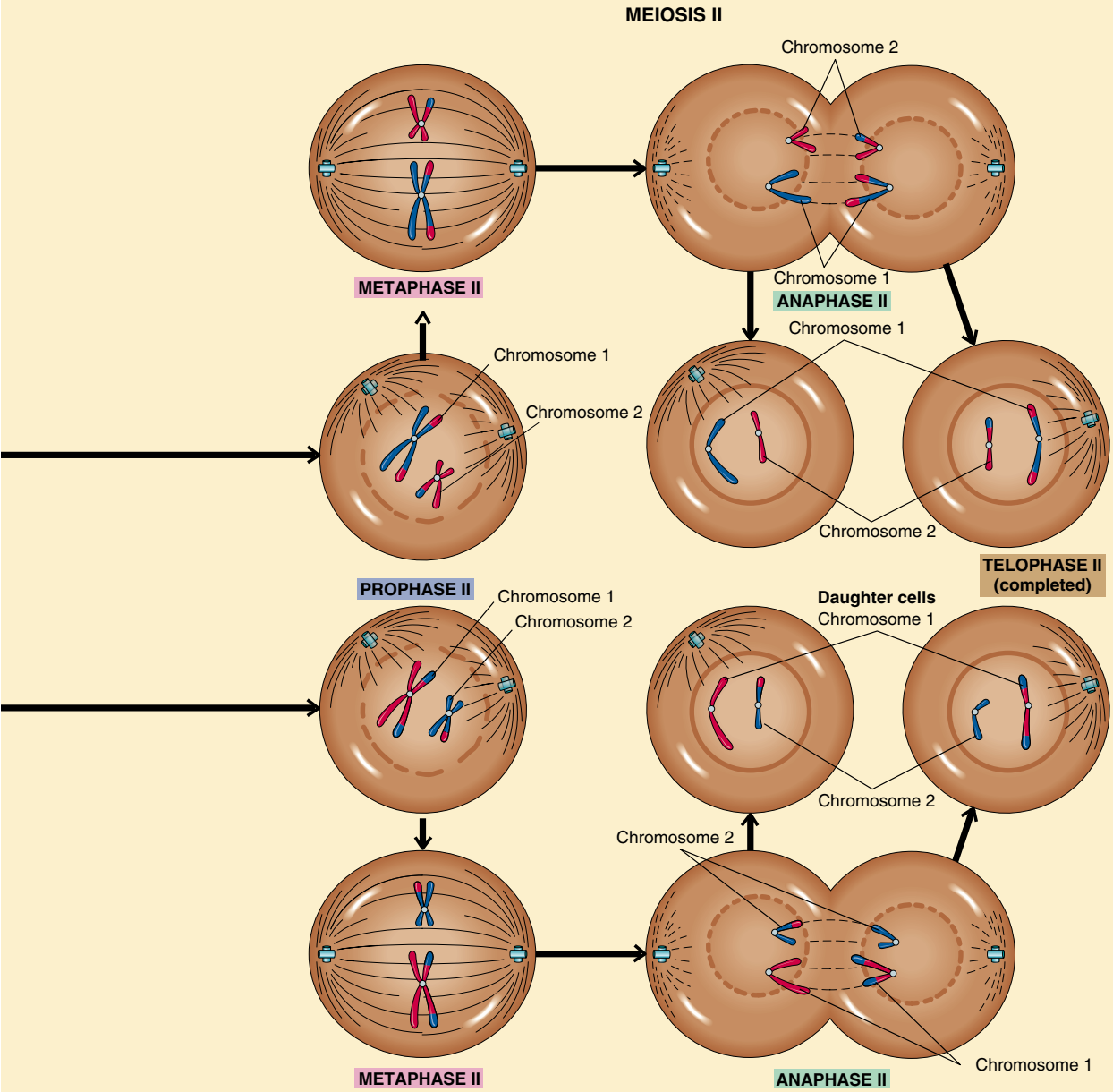




Figure 1.5 Barbara McClintock. (*Source:* Bettmann Archive.)

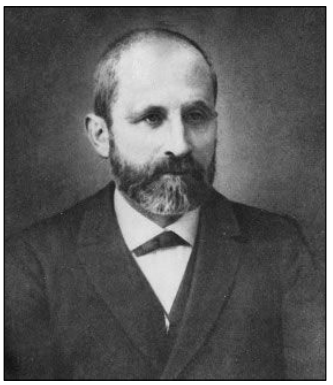


Figure 1.6 Friedrich Miescher. (*Source:* National Library of Medicine.)

Physical Evidence for Recombination

Barbara McClintock (Figure 1.5) and Harriet Creighton provided a direct physical demonstration of recombination in 1931. By examining maize chromosomes microscopically, they could detect recombinations between two easily identifiable features of a particular chromosome (a knob at one end and a long extension at the other). Furthermore, whenever this physical recombination occurred, they could also detect recombination genetically. Thus, they established a direct relationship between a region of a chromosome and a gene. Shortly after McClintock and Creighton performed this work on maize, Curt Stern observed the same phenomenon in *Drosophila*. So recombination could be detected both physically and genetically in animals as well as plants.

SUMMARY The chromosome theory of inheritance holds that genes are arranged in linear fashion on chromosomes. The reason that certain traits tend to be inherited together is that the genes governing these traits are on the same chromosome. However, recombination between two chromosomes during meiosis can scramble the parental alleles to give nonparental combinations. The farther apart two genes are on a chromosome the more likely such recombination between them will be.

1.2 Molecular Genetics

The studies just discussed tell us important things about the transmission of genes and even about how to map genes on chromosomes, but they do not tell us what genes are made of or how they work. This has been the province of molecular genetics, which also happens to have its roots in Mendel's era.

The Discovery of DNA

In 1869, Friedrich Miescher (Figure 1.6) discovered in the cell nucleus a mixture of compounds that he called nuclein. The major component of nuclein is **deoxyribonucleic acid (DNA).** By the end of the 19th century, chemists had learned the general structure of DNA and of a related compound, **ribonucleic acid (RNA).** Both are long polymers—chains of small compounds called nucleotides. Each nucleotide is composed of a sugar, a phosphate group, and a base. The chain is formed by linking the sugars to one another through their phosphate groups.

The Composition of Genes

By the time the chromosome theory of inheritance was generally accepted, geneticists agreed that the chromosome must be composed of a polymer of some kind. This would agree with its role as a string of genes. But which polymer is it? Essentially, the choices were three: DNA, RNA, and **protein.** Protein was the other major component of Miescher's nuclein; its chain is composed of links called **amino acids.** The amino acids in protein are joined by **peptide bonds,** so a single protein chain is called a **polypeptide.**

Oswald Avery (Figure 1.7) and his colleagues demonstrated in 1944 that DNA is the right choice (see Chapter 2). These investigators built on an experiment performed earlier by Frederick Griffith in which he transferred a genetic trait from one strain of bacteria to another. The trait was virulence, the ability to cause a lethal infection, and it could be transferred simply by mixing dead virulent cells with live avirulent (nonlethal) cells. It was very likely that the substance that caused the transformation from avirulence to virulence in the recipient cells was the gene for virulence, because the recipient cells passed this trait on to their progeny.

What remained was to learn the chemical nature of the transforming agent in the dead virulent cells. Avery

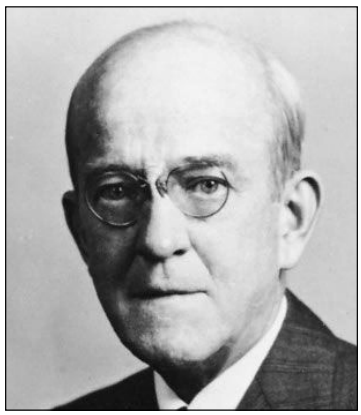
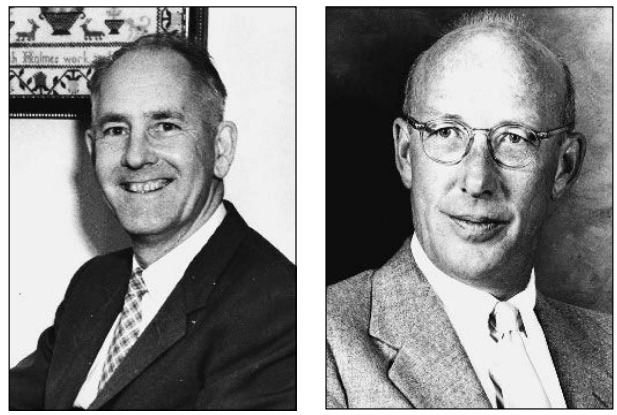


Figure 1.7 Oswald Avery. (*Source:* National Academy of Sciences.)



(a) (b)

Figure 1.8 (a) George Beadle; (b) E.L. Tatum. (*Source: (a, b)* AP/Wide World Photos.)

and his coworkers did this by applying a number of chemical and biochemical tests to the transforming agent, showing that it behaved just as DNA should, not as RNA or protein should.

The Relationship between Genes and Proteins

The other major question in molecular genetics is this: How do genes work? To lay the groundwork for the answer to this question, we have to backtrack again, this time to 1902. That was the year Archibald Garrod noticed that the human disease alcaptonuria seemed to behave as a Mendelian recessive trait. It was likely, therefore, that the disease was caused by a defective, or mutant, gene. Moreover, the main symptom of the disease was the accumulation of a black pigment in the patient's urine, which Garrod believed derived from the abnormal buildup of an intermediate compound in a biochemical pathway.

By this time, biochemists had shown that all living things carry out countless chemical reactions and that these reactions are accelerated, or catalyzed, by proteins called **enzymes.** Many of these reactions take place in sequence, so that one chemical product becomes the starting material, or substrate, for the next reaction. Such sequences of reactions are called **pathways,** and the products or substrates within a pathway are called **intermediates.** Garrod postulated that an intermediate accumulated to abnormally high levels in alcaptonuria because the enzyme that would normally convert this intermediate to the next was defective. Putting this idea together with the finding that alcaptonuria behaved genetically as a Mendelian recessive trait, Garrod suggested that a defective gene gives rise to a defective enzyme. To put it another way: A gene is responsible for the production of an enzyme.

Garrod's conclusion was based in part on conjecture; he did not really know that a defective enzyme was involved in alcaptonuria. It was left for George Beadle and E. L. Tatum (Figure 1.8) to prove the relationship between genes and enzymes. They did this using the mold *Neurospora* as their experimental system. *Neurospora* has an enormous advantage over the human being as the subject of genetic experiments. By using *Neurospora,* scientists are not limited to the mutations that nature provides, but can use **mutagens** to introduce mutations into genes and then observe the effects of these mutations on biochemical pathways. Beadle and Tatum found many instances where they could create *Neurospora* mutants and then pin the defect down to a single step in a biochemical pathway, and therefore to a single enzyme (see Chapter 3). They did this by adding the intermediate that would normally be made by the defective enzyme and showing that it restored normal growth. By circumventing the blockade, they discovered where it was. In these same cases, their genetic experiments showed that a single gene was involved. Therefore, a defective gene gives a defective (or absent) enzyme. In other words, a gene seemed to be responsible for making one enzyme. This was the one-gene/one-enzyme hypothesis. This hypothesis was actually not quite right for at least three reasons: (1) An enzyme can be composed of more than one polypeptide chain, whereas a gene has the information for making only one polypeptide chain. (2) Many genes contain the information for making polypeptides that are not enzymes. (3) As we will see, the end products of some genes are not polypeptides, but RNAs. A modern restatement of the one-gene/one-enzyme hypothesis would be: Most genes contain the information for making one polypeptide.

Activities of Genes

Let us now return to the question at hand: How do genes work? This is really more than one question because genes do more than one thing. First, they are replicated

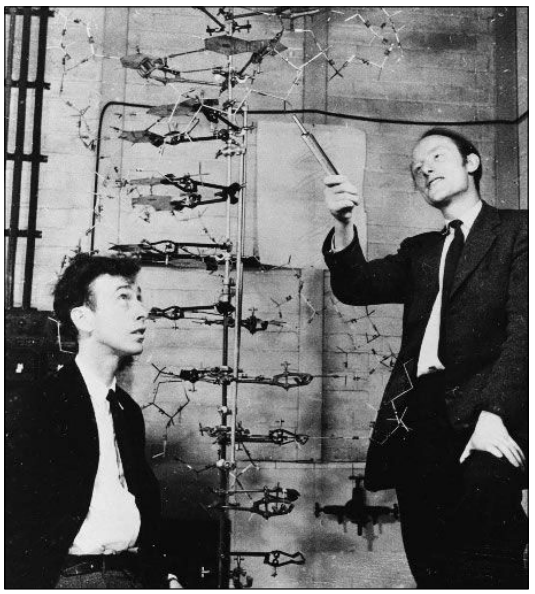
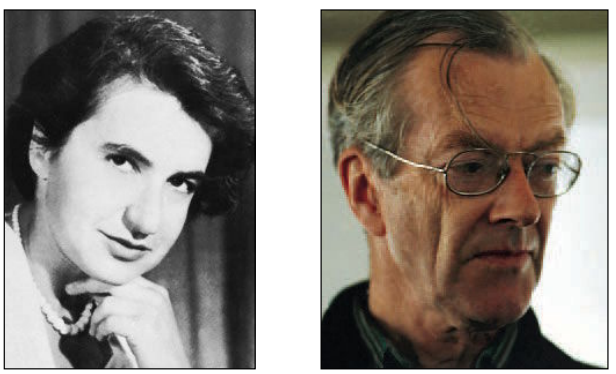


Figure 1.9 James Watson (left) and Francis Crick. (*Source:* Corbis/Bettmann Archive.)

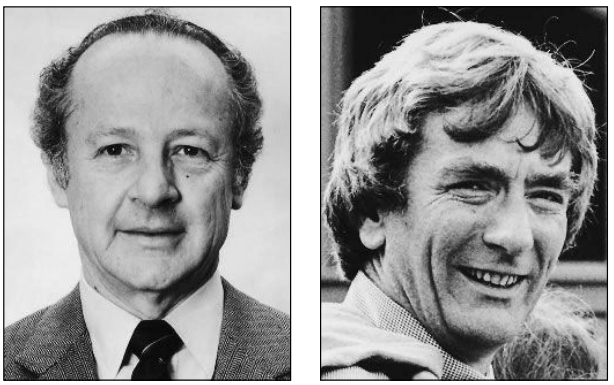


(a) (b)

Figure 1.10 (a) Rosalind Franklin; (b) Maurice Wilkins. (*Source:* (*a*) From *The Double Helix* by James D. Watson, 1968, Atheneum Press, NY. © Cold Spring Harbor Laboratory Archives. (*b*) Courtesy Professor M. H. F. Wilkins, Biophysics Dept., King's College, London.)

faithfully; second, they direct the production of RNAs and proteins; third, they accumulate mutations and so allow evolution. Let us look briefly at each of these activities.

How Genes Are Replicated First of all, how is DNA replicated faithfully? To answer that question, we need to know the overall structure of the DNA molecule as it is found in the chromosome. James Watson and Francis Crick (Figure 1.9) provided the answer in 1953 by building models based on chemical and physical data that had been gathered in other laboratories, primarily x-ray diffraction data collected by Rosalind Franklin and Maurice Wilkins (Figure 1.10).



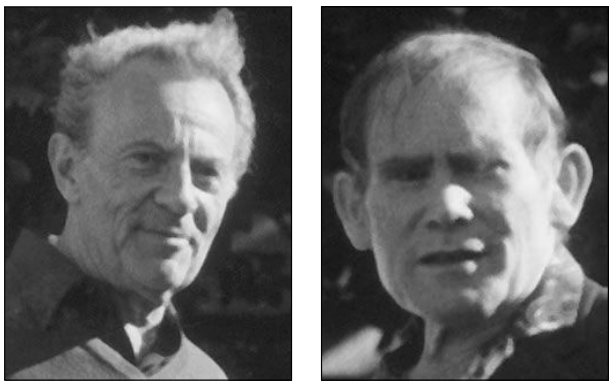
(a) (b)

Figure 1.11 (a) Matthew Meselson; (b) Franklin Stahl. (*Source:* (*a*) Courtesy Dr. Matthew Meselson. (*b*) Cold Spring Harbor Laboratory Archives.)

Watson and Crick proposed that DNA is a **double helix**—two DNA strands wound around each other. More important, the bases of each strand are on the inside of the helix, and a base on one strand pairs with one on the other in a very specific way. DNA has only four different bases: adenine, guanine, cytosine, and thymine, which we abbreviate A, G, C, and T. Wherever we find an A in one strand, we always find a T in the other; wherever we find a G in one strand, we always find a C in the other. In a sense, then, the two strands are complementary. If we know the base sequence of one, we automatically know the sequence of the other. This complementarity is what allows DNA to be replicated faithfully. The two strands come apart, and enzymes build new partners for them using the old strands as templates and following the Watson–Crick base-pairing rules (A with T, G with C). This is called **semiconservative replication** because one strand of the parental double helix is conserved in each of the daughter double helices. In 1958, Matthew Meselson and Franklin Stahl (Figure 1.11) proved that DNA replication in bacteria follows the semiconservative pathway (see Chapter 20).

How Genes Direct the Production of Polypeptides **Gene expression** is the process by which a gene product (an RNA or a polypeptide) is made. Two steps, called **transcription** and **translation,** are required to make a polypeptide from the instructions in a DNA gene. In the transcription step, an enzyme called RNA polymerase makes a copy of one of the DNA strands; this copy is not DNA, but its close cousin RNA. In the translation step, this RNA (**messenger RNA,** or **mRNA**) carries the genetic instructions to the cell's protein factories, called **ribosomes.** The ribosomes "read" the **genetic code** in the mRNA and put together a protein according to its instructions.

Actually, the ribosomes already contain molecules of RNA, called **ribosomal RNA** (**rRNA**). Francis Crick originally thought that this RNA residing in the ribosomes



(a) (b)

Figure 1.12 (a) François Jacob; (b) Sydney Brenner. (*Source: (a, b)* Cold Spring Harbor Laboratory Archives.)

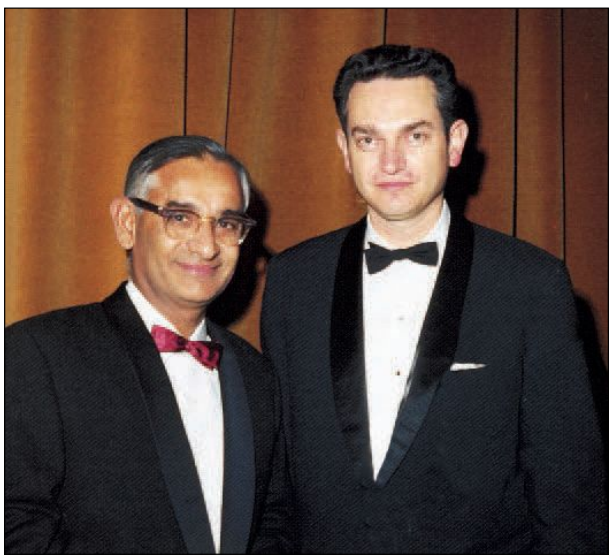


Figure 1.13 Gobind Khorana (left) and Marshall Nirenberg. (*Source:* Corbis/Bettmann Archive.)

carried the message from the gene. According to this theory, each ribosome would be capable of making only one kind of protein—the one encoded in its rRNA. François Jacob and Sydney Brenner (Figure 1.12) had another idea: The ribosomes are nonspecific translation machines that can make an unlimited number of different proteins, according to the instructions in the mRNAs that visit the ribosomes. Experiment has shown that this idea is correct (see Chapter 3).

What is the nature of this genetic code? Marshall Nirenberg and Gobind Khorana (Figure 1.13), working independently with different approaches, cracked the code in the early 1960s (see Chapter 18). They found that 3 bases constitute a code word, called a **codon,** that stands for one amino acid. Out of the 64 possible 3-base codons, 61 specify amino acids; the other three are stop signals. The ribosomes scan a messenger RNA 3 bases at a time and bring in the corresponding amino acids to link to the growing protein chain. When they reach a stop signal, they release the completed protein.

How Genes Accumulate Mutations Genes change in a number of ways. The simplest is a change of one base to another. For example, if a certain codon in a gene is GAG (for the amino acid called glutamate), a change to GTG converts it to a codon for another amino acid, valine. The protein that results from this mutated gene will have a valine where it ought to have a glutamate. This may be one change out of hundreds of amino acids, but it can have profound effects. In fact, this specific change has occurred in the gene for one of the human blood proteins and is responsible for the genetic disorder we call sickle cell disease.

Genes can suffer more profound changes, such as deletions or insertions of large pieces of DNA. Segments of DNA can even move from one locus to another. The more drastic the change, the more likely that the gene or genes involved will be totally inactivated.

Gene Cloning In recent years, geneticists have learned to isolate genes, place them in new organisms, and reproduce them by a set of techniques collectively known as **gene cloning.** Cloned genes not only give molecular biologists plenty of raw materials for their studies, they also can be induced to yield their protein products. Some of these, such as human insulin or blood clotting factors, can be very useful. Cloned genes can also be transplanted to plants and animals, including humans. These transplanted genes can alter the characteristics of the recipient organisms, so they may provide powerful tools for agriculture and for intervening in human genetic diseases. We will examine gene cloning in detail in Chapter 4.

SUMMARY Most genes are made of DNA arranged in a double helix. This structure explains how genes engage in their three main activities: replication, carrying information, and collecting mutations. The complementary nature of the two DNA strands in a gene allows them to be replicated faithfully by separating and serving as templates for the assembly of two new complementary strands. The sequence of nucleotides in a typical gene is a genetic code that carries the information for making an RNA. Most of these are messenger RNAs that carry the information to protein-synthesizing ribosomes. The end result is a new protein chain made according to the gene's instructions. A change in the sequence of bases constitutes a mutation, which can change the sequence of amino acids in the gene's protein product. Genes can be cloned, allowing molecular biologists to harvest abundant supplies of their products.

Table 1.1 Molecular Biology Time Line

1859	Charles Darwin	Published *On the Origin of Species*
1865	Gregor Mendel	Advanced the principles of segregation and independent assortment
1869	Friedrich Miescher	Discovered DNA
1900	Hugo de Vries, Carl Correns, Erich von Tschermak	Rediscovered Mendel's principles
1902	Archibald Garrod	First suggested a genetic cause for a human disease
1902	Walter Sutton, Theodor Boveri	Proposed the chromosome theory
1908	G.H. Hardy, Wilhelm Weinberg	Formulated the Hardy–Weinberg principle
1910, 1916	Thomas Hunt Morgan, Calvin Bridges	Demonstrated that genes are on chromosomes
1913	A.H. Sturtevant	Constructed a genetic map
1927	H.J. Muller	Induced mutation by x-rays
1931	Harriet Creighton, Barbara McClintock	Obtained physical evidence for recombination
1941	George Beadle, E.L. Tatum	Proposed the one-gene/one-enzyme hypothesis
1944	Oswald Avery, Colin McLeod, Maclyn McCarty	Identified DNA as the material genes are made of
1953	James Watson, Francis Crick, Rosalind Franklin, Maurice Wilkins	Determined the structure of DNA
1958	Matthew Meselson, Franklin Stahl	Demonstrated the semiconservative replication of DNA
1961	Sydney Brenner, François Jacob, Matthew Meselson	Discovered messenger RNA
1966	Marshall Nirenberg, Gobind Khorana	Finished unraveling the genetic code
1970	Hamilton Smith	Discovered restriction enzymes that cut DNA at specific sites, which made cutting and pasting DNA easy, thus facilitating DNA cloning
1972	Paul Berg	Made the first recombinant DNA in vitro
1973	Herb Boyer, Stanley Cohen	First used a plasmid to clone DNA
1977	Walter Gilbert and Frederick Sanger	Worked out methods to determine the sequence of bases in DNA
1977	Frederick Sanger	Determined the base sequence of an entire viral genome (ϕX174)
1977	Phillip Sharp, Richard Roberts, and others	Discovered interruptions (introns) in genes
1990	Lap-Chee Tsui, Francis Collins, John Riordan	Found the gene that is responsible for cystic fibrosis
1990	James Watson and many others	Launched the Human Genome Project to map the entire human genome and, ultimately, to determine its base sequence
1993	The Huntington's Disease Group	Identified the Huntington's disease gene
1995	Craig Venter, Hamilton Smith	Determined the base sequences of the genomes of two bacteria: *Hemophilus influenzae* and *Mycoplasma genitalium,* the first genomes of free-living organisms to be sequenced
1996	Many investigators	Determined the base sequence of the genome of brewer's yeast, *Saccharomyces cerevisiae,* the first eukaryotic genome to be sequenced
1997	Fred Blattner, Takashi Honuchi and many colleagues	Determined the base sequence of the genome of *Escherichia coli*
1997	Ian Wilmut and colleagues	Cloned a sheep (Dolly) from an adult sheep udder cell
1998	Many investigators	Determined the base sequence of the genome of the roundworm *Caenorhabditis elegans,* the first animal genome to be sequenced
1999	Many investigators	Determined the base sequence of human chromosome 22
2000	Many investigators	Determined the base sequence of the genome of the fruit fly, *Drosophila melanogaster,* a mainstay of genetic research
2000	Many investigators	Determined the base sequence of the genome of the mustard, *Arabidopsis thaliana,* the first plant genome to be sequenced
2000	Alain Fischer and colleagues	First clearly successful gene therapy trial—on two patients with severe combined immunodeficiency ("bubble boy" syndrome)
2001	Craig Venter, Francis Collins, and many others	Reported working draft of human genome sequence

This concludes our brief chronology of molecular biology. Table 1.1 reviews some of the milestones. Although it is a very young discipline, it has an exceptionally rich history, and molecular biologists are now adding new knowledge at an explosive rate. Indeed, the pace of discovery in molecular biology, and the power of its techniques, has led many commentators to call it a revolution. Because some of the most important changes in medicine and agriculture over the next few decades are likely to depend on the manipulation of genes by molecular biologists, this revolution will touch everyone's life in one way or another. Thus, you are embarking on a study of a subject that is not only fascinating and elegant, but one that has practical importance as well. F.H. Westheimer, professor emeritus of chemistry at Harvard University put it well: "The greatest intellectual revolution of the last 40 years may have taken place in biology. Can anyone be considered educated today who does not understand a little about molecular biology?" Happily, after this course you should understand more than a little.

SUMMARY

Genes can exist in several different forms called alleles. A recessive allele can be masked by a dominant one in a heterozygote, but it does not disappear. It can be expressed again in a homozygote bearing two recessive alleles.

Genes exist in a linear array on chromosomes. Therefore, traits governed by genes that lie on the same chromosome can be inherited together. However, recombinations between homologous chromosomes occur during meiosis, so that gametes bearing nonparental combinations of alleles can be produced. The farther apart two genes lie on a chromosome, the more likely such recombination between them will be.

Most genes are made of double-stranded DNA arranged in a double helix. One strand is the complement of the other, which means that faithful gene replication requires that the two strands separate and acquire complementary partners. The linear sequence of bases in a typical gene carries the information for making a protein.

The process of making a gene product is called gene expression. It occurs in two steps: transcription and translation. In the transcription step, RNA polymerase makes a messenger RNA, which is a copy of the information in the gene. In the translation step, ribosomes "read" the mRNA and make a protein according to its instructions. Thus, a change (mutation) in a gene's sequence may cause a corresponding change in the protein product.

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